
Neutrinos via Charm Decays in Astrophysical Sources

TALK IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE DEGREE OF BACHELOR OF SCIENCE

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Particle Physics

- Point A
- Point B
 - Part 1
 - Part 2
- Point C
- Point D



Multimessenger Astronomy

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Astrophysical Sources

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Cross Sections

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Spectral Distributions

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Computation

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Results

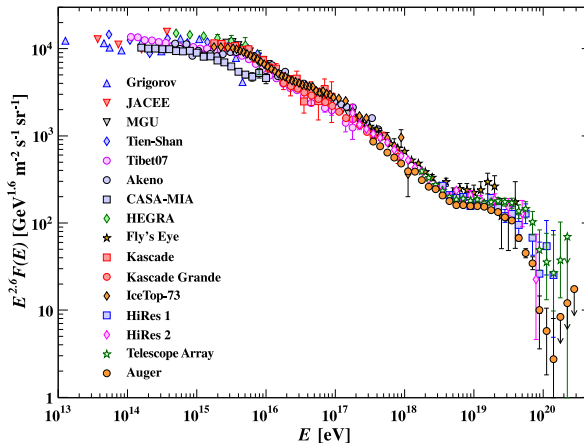
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Conclusion & Outlook

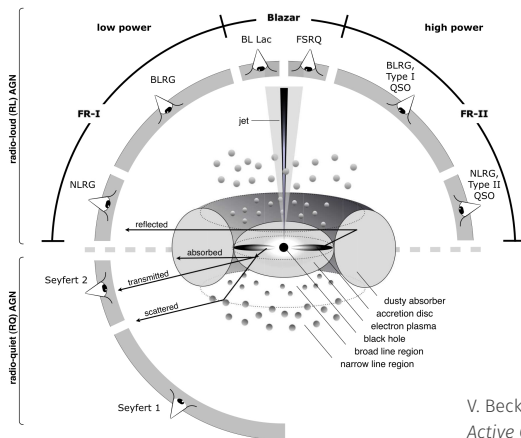
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Cosmic Ray Spectrum & Cutoff



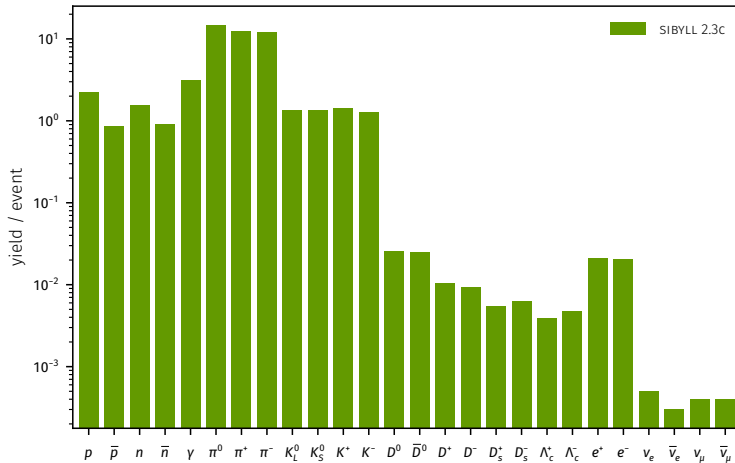
K. A. Olive et. al.
Chin. Phys. C (2014)

Unified AGN Classification Model



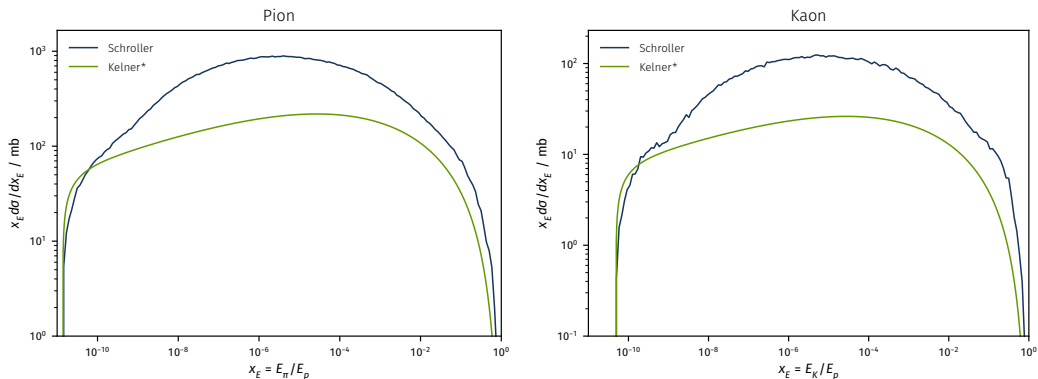
V. Beckmann, C. Shrader
Active Galactic Nuclei (2012)

Event Generation



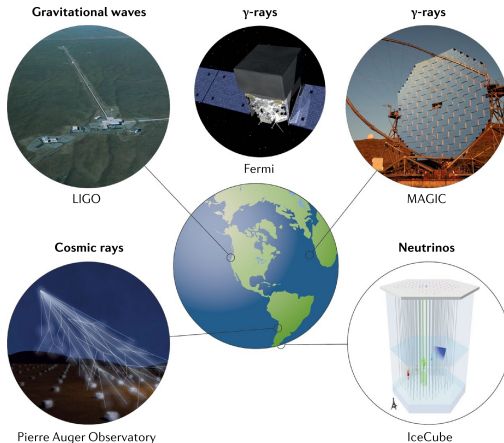
Data Courtesy
M. Schroller

Weighted Energy Pion & Kaon Differential Cross Section



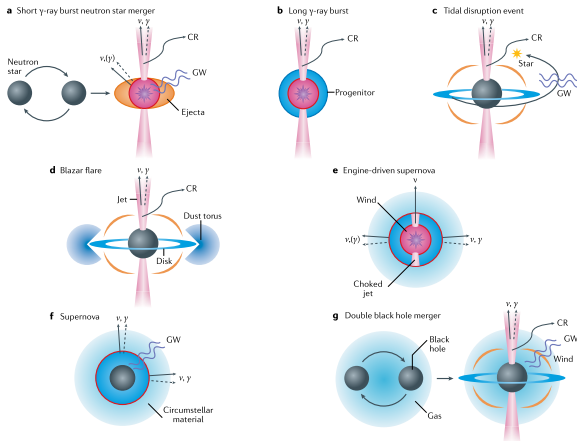
Derived Parametrization S. R. Kelner et. al. *Phys. Rev. D* (2006)
Event Data (SIBYLL 2.3c) Courtesy M. Schrolller

Detectors



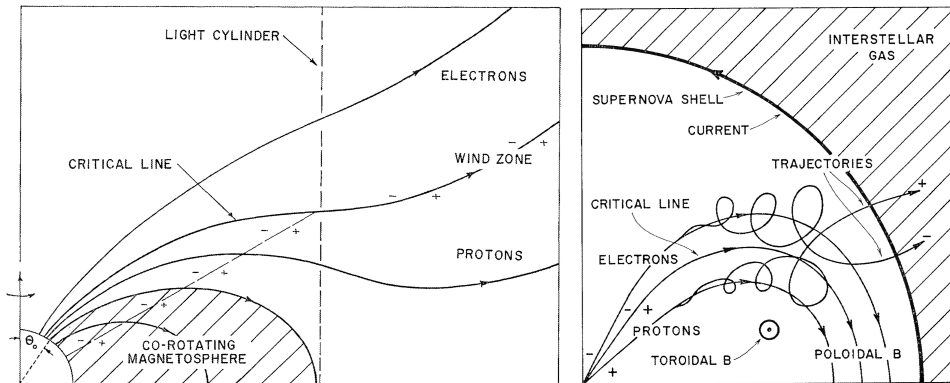
P. Mészáros et. al.
Nat. Rev. Phys. (2019)

Sources



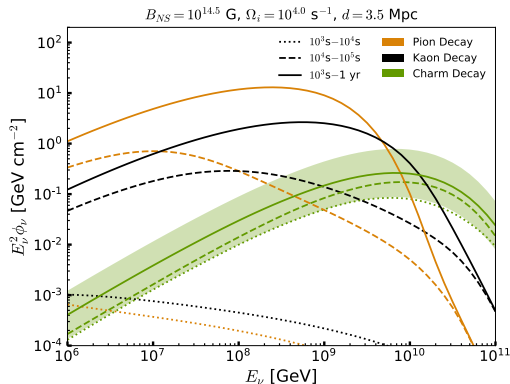
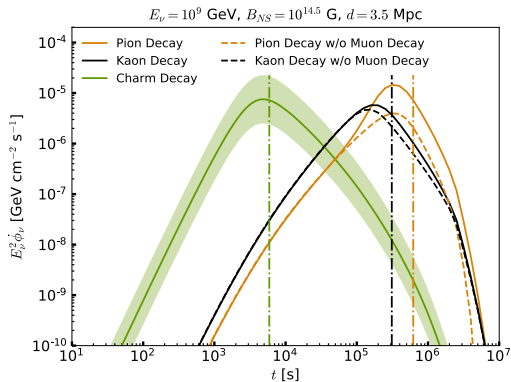
P. Mészáros et. al.
Nat. Rev. Phys. (2019)

Pulsar Particle Acceleration



P. Goldreich, W. H. Julian *ApJ* (1969)

Magnetar Neutrino Flux & Fluence Comparison



J. A. Carpio et. al. Phys. Rev. D (2020)